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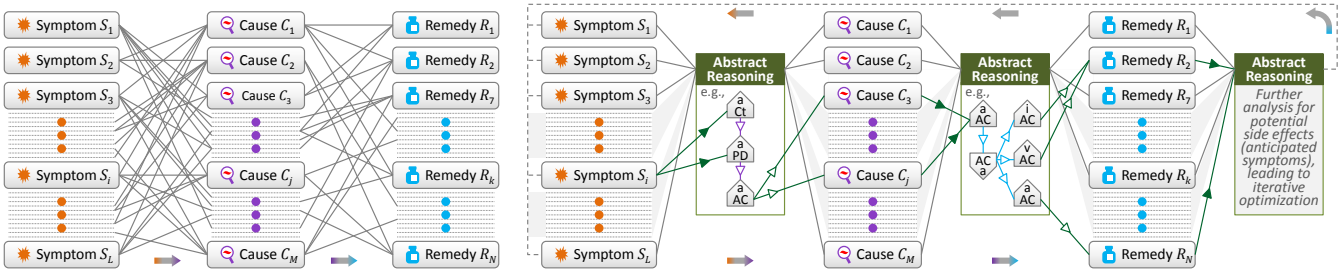
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An Ontological Framework for Supporting the Design and Evaluation of Visual Analytics Systems

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(a) an ontology with concrete entities only (b) an ontology with both abstract and concrete entities

Figure 1: *An ontology can support the design and evaluation of visual analytics systems by enabling systematic recording of symptoms, causes, remedies, and side-effects, facilitating the analysis of causal relations, search for solutions, and comparison of different design options. (a) An ontology that records only concrete entities (e.g., instances in practice) will take years and decades to build. (b) An ontology built around abstract entities can enable abstract reasoning in a scalable manner while allowing gradual addition of concrete entities.*

Abstract

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. In this paper, we present an ontological framework for recording and categorizing technical shortcomings to be addressed in a VA workflow, reasoning about the causes of such problems, identifying technical solutions, and anticipating secondary effects of the solutions. The methodology is built on the theoretical premise that designing a VA workflow is an optimization of the cost-benefit ratio of the processes in the workflow. It makes uses three fundamental measures to group and connect "symptoms", "causes", "remedies", and "side-effects", and guide the search for potential solutions to the problems. In terms of requirement analysis and system design, the proposed methodology can enable system designers to explore the decision space in a structured manner. In terms of evaluation, the proposed methodology is time-efficient and complementary to various forms of empirical studies, such as user surveys, controlled experiments, observational studies, focus group discussions, and so on. In general, it reduces the amount of trial-and-error in the lifecycle of VA system development.

1. Introduction

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. Since the first IEEE VAST in 2006, a large number of system papers and application papers have been published, signifying the important contributions to the technology advancements as well as a variety of successful applications [WHC*17]. During this relatively short period, VA researchers have made noticeable progress in adopting some best practices in software engineering (e.g., [Sum15]). Today, most system papers and application papers in VA conferences reported user-centered requirement analysis and evaluation. Some reported the use of incremental, iterative, nested, agile, and evolutionary development approaches [Lar03, Mun09, Mar14].

While these approaches have no doubt successfully delivered many VA systems and applications, assessing and sharing the experience of developing these VA systems and applications can be difficult. These difficulties include:

- The step made from requirements analysis to design specification is typically based on intuitions and the design decisions made are often undocumented, difficult to share, or not reproducible in another application.
- The evaluation based on individual use cases is sometimes per-

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ceived to be anecdote evidence and may be biased towards positive conclusions.
- The evaluation based on users' testimonies is usually difficult to verify and reconfirm in another application environment.
- The readers (including reviewers) are often left with the options to either take the reported design and evaluation passively or make speculative comments such as "*I am not convinced*" without substantiated reasoning.
- In some application areas, user-centered evaluation is often hindered by various difficulties such as data confidentiality and deployment delays (e.g., due to the necessity for changing the current working practice or training users).

An *ontology*, which is a form of knowledge representation, typically consists of a collection of entities (including concepts and categories) in a subject domain, relationships between entities, and connections from entities to their properties. Ideally, there could be an ontology that captures the experience and knowledge of VA researchers and developers by recording and categorizing various technical shortcomings to be addressed in a VA workflow, their analysis about the causes of such problems, the design options of possible technical solutions, the undesirable secondary effects due to the chosen solutions, the further solutions for addressing the secondary effects, and so forth. In many ways, this bears some resemblance to the notions of "symptoms", "causes", "remedies", and "side-effects" in medicine [WRW*17,JRW*18,TO17], which have been studied and documented for centuries and millennia.

In this paper, we propose an ontological framework, as the first step to build such an ontology gradually for supporting the design and evaluation of VA systems and applications. While it may be possible to collect a comprehensive list of "symptoms", "causes", "remedies", and "side-effects" in VA workflows after years or decades of effort, it is important to establish a set of abstract categories to serve as the core of the ontology. Under this framework, each instance of "symptoms", "causes", "remedies", and "side-effects" observed in practice is mapped to one or a few abstract categories. This facilitates abstract reasoning about the causal relations in a much smaller problem-solution space and in a scalable manner.

As illustrated in Figure 1(a), an ontology based only on concrete entities (i.e., practical instances) will takes years or decades to mature, and a sparsely filled ontology will be unsafe to use. A VA practitioner could easily be overwhelmed by the rapid expansion of the options to be considered, and be rather anxious about many options that might have not been recorded in the ontology.

Alternatively, if an evolving ontology can be built around a stable set of abstract entities, as shown in Figure 1(b), VA practitioners can conduct abstract reasoning systematically, while drawing knowledge from recorded concrete entities as well as using their own experience to identify new instances of "symptoms", "causes", "remedies", and "side-effects" in different applications.

Our abstract categories are defined based on two types of properties, namely *four primary components* and *three fundamental measures*. The former relates a practical instance to different machine- and human-centric processes in VA workflows, such as statistics, algorithms, visualization, and interaction [CTBA*11]. The latter relates a practical instance to different information-theoretic measures in the cost-benefit metric for optimizing VA processes [CG16]. For abstract reasoning in design optimization, we consider two states ("high" and "low") of the three measures. We have thus a total of 24 abstract entities, which define a much smaller problem-solution space for enumerating a reasonably complete set of causal relationships among these entities. While more concrete entities can gradually be added, the ontological framework can start to provide support to VA practitioners in their analyzing the shortcomings of a current VA workflow, cogitating likely causes, deliberating possible solutions, and anticipating undesirable secondary effects in a structured manner.

In terms of evaluation, the analytical methodology based on the proposed ontological framework is complementary to various forms of empirical methods, such as user surveys, controlled experiments, observational studies, focus group discussions, and so on. In general, it reduces the amount of trial-and-error in the developmental lifecycles of VA systems, and facilitates the recording, sharing, and reproduction of high-level design knowledge, while empowering third parties (e.g., readers and reviewers) to verify the design rationales analytically. We recognize that the abstract categorization and practical examples of symptoms, causes, and remedies in this work are only a starting point. Building a comprehensive ontology as a knowledge base for VA practitioners will be a long-term and collective endeavor of the visualization community, and many new symptoms, causes, and remedies will continue to emerge as part of technical innovation, scientific investigation, and practical application.

2. Related Work

Since the term *Visual Analytics* (VA) was coined in 2004 [WT04, TC05], there have been hundreds of software systems reported in the literature for supporting various VA workflows (e.g., [ERT*17, LGH*17,CLY17]). As articulated by Thomas and Kielman [TK09] and Robertson et. al. [REE*09], it has always been a challenging undertaking to design, develop, evaluate, and improve VA systems, because of the needs to integrate a range of technologies (e.g., data mining, machine learning, visualization, and human-computer interaction), to respond to the rapid changes of the environments where data is captured, to take advantage of users' knowledge and experience while addressing their cognitive limitations, and often to support mission-critical operations such as emergency and healthcare services. In the VA literature, such challenges have stimulated scholarly discourses on VA workflows [KAF*08], VA processes [KS11], ensemble tools usage [ALSS07], cognition [GRF08], evaluation [Sch06], collaboration [HA07], insight provenance [GZ08], and so on. This work examines an abstract reasoning methodology behind those discourses on practical design strategies for optimizing different aspects of VA systems.

Design and Evaluation Methods for Visualization and VA. Abram and Treinish proposed to implement visualization processes on data-flow architectures [AT95]. Chi described visualization processes using a state reference model, involving data, visualization, and visual mapping transformation [Chi00]. Jansen and Dragicevic proposed an interaction model in the context of visualization pipelines [JD13]. Munzner proposed a nested model for designing and developing visualization pipelines [Mun09]. Brehmer

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and Munzner proposed a multi-level typology for task abstraction [BM13], Wang et al. proposed a two-stage framework for designing visual analytics systems [WDB*11]. Kandel et al. categorized VA workflows based on an interview study [KPHH12]. Ahmed et al. proposed to use purpose-driven games for evaluating visualization systems [AZM12]. Scholtz outlined a set of guidelines for assessing visual analytics environments [Sch11]. Scholtz et al. further developed them into an evaluation methodology [SPWG13]. von Landesberger et al. considered a categorization of system requirements for data processing in visualization [vFR17]. Lam et al. analyzed the relationships between goals and tasks in 20 visualization design studies [LTM18]. The ontological framework proposed in this work can be considered as a new scheme for categorizing (or coding) the problems and solutions identified or considered in designing, developing, evaluating, and optimizing VA processes, systems, and workflows. In particular, this new scheme enables theory-guided reasoning about the relationships between problems and solutions.

Many researchers have discussed the design spaces for visualization (e.g., [CM97, SHS11, JE12, SNHS13, CBL*17, VZv*17]). More than a decade ago, Chen advocated the need for design patterns from the perspective of software architectures [Che04]. Tory and Moller examined the design patterns from data to visual representations [TM04]. Heer and Agrawala described 12 design patterns under an object-oriented framework [HA06]. Giereth and Ertl introduced three further design patterns [GE08]. Since then, there have been a few attempts to develop and use design patterns for visualization (e.g., [HLZY10, PSDK15]). One major stumbling block is that there will be a huge list of design patterns that record solutions for different problems, considering the complexity of the design space of VA workflows [TK09, REE*09]. This work attempts to tackle this challenge in a scalable manner.

The complexity of the VA design space is comparable to various relational patterns in other disciplines, such as symptom, cause, and treatment in medicine [Ngu00, Eis02, Leo11], requirement and design in architecture [AIS*77], employee and career development in social science [GM91], management problem and solution in businesses [RSP*00], design problem and solution in software engineering [GHJV94, Fow96], and so on. This work addresses the complexity by considering a highly abstracted design space, enumerating the design patterns within this abstract space, and initiating a longer term research effort for compiling a relatively comprehensive collection of design patterns in the form of an ontology for supporting the design, evaluation, and improvement of VA systems.

Theoretical Foundation. As parts of the development life-cycle of a VA system, design, evaluation, and improvement are all means for enabling the VA system to provide optimal support to the users concerned and their workflows. The information-theoretic metric for analyzing the cost-benefit ratio of VA workflows [CG16] is naturally a candidate for providing the basis of such optimization efforts. The cost-benefit metric was recently used to analyze a wide range of visualization tasks in virtual environments (e.g., virtual reality, mixed reality, and non-immersive environments) [CGJM19], demonstrating its applicability to practical scenarios.

Taxonomies and ontologies play a major role in the theoretical foundation of visualization and VA [CGJ*17]. In particular, a collection of problem-solution patterns is a collection of knowledge, and an ontology is a form of knowledge base. There have been several previous attempts to create ontologies for different aspects of visualization [BDD04, DBD04, DBDH05, SAR06, GSGC08, PRG10, PV13, VP11] and data mining [PDS08, PSD09, PSD13, KPD15, SS16]. Sacha et al. recently developed the first ontology in the area of VA, covering the aspect of VA-assisted machine learning [SKKC19]. Built on these theoretical developments, this work aims to bridge the gap between theory and practice.

3. From Practical Instances to Theoretical Abstraction

The design space for VA systems is a high-dimensional design space. In practice, there are normally many aspects to be considered by VA practitioners. Ideally, one could decompose this massive design space into smaller sections, where known design patterns are recorded and searchable. In many ways, VA practitioners would benefit from accessing some "well-organized medicine cabinets" for potential solutions. For user experience analysts, ideally, one could conduct a good portion of assessment through theory-guided reasoning and identify likely shortcomings for further user-centered evaluation.

To decompose this massive, complicated design problem into many smaller problems that can be tackled progressively, we first make the following assumptions:

(a) We assume that a VA system being designed or evaluated supports a VA workflow, which may involve multiple analytical activities. The workflow can be decomposed into a series of machine- and human-centric processes, $P_1, P_2, \ldots, P_n$. Any parallel processes in practice can either be sequentialized mathematically or be merged into a single process.
(b) We assume that the design of a VA system will follow an agile and iterative approach as advocated in [Mun09]. With this in mind, there will always be at least one *reference design*, with which to compare a new design. The reference design may be in the form of an existing VA system currently in operation but considered not ideal, a VA system reported in the literature, or even an existing VA workflow without any software support.
(c) With (a) and (b), the design of a VA system can be seen as a series of actions for identifying problems in different processes of a VA workflow, considering the possible causes, deliberating optional remedies, and determining the optimal solution.
(d) Similarly, the evaluation of a new VA system (or its design specification) can be seen as considering the relative merits and demerits that a new solution may engender. If the demerits are confirmed, one may start a new iteration of (c) and (d).

Metaphorically, we can consider the transition from "a problem" to "a cause" and then to a "solution" as a pattern similar to "a symptom" to a "cause" and then to a "remedy" in medicine. In evaluation, following similar metaphors, we can consider a "merit" as a "beneficial effect", and a "demerit" as an "adverse side-effect".

3.1. Symptoms, Causes, Remedies, and Side-Effects

Consider two commonly-encountered symptoms:

S_1 A current human-centric decision process P_i is too slow.
S_2 A current machine-centric decision process P_j is unreliable.

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There may be many causes that may lead to S_1 and S_2. For example, S_1 may be caused by the following shortcomings in P_i or its preceding processes:

$C_{1,1}$ A poorly designed user interface may cause slow interaction processes.
$C_{1,2}$ Too much data may cause slow visualization processes.
$C_{1,3}$ Some visual representations may incur high cognitive load, causing slow visualization processes.
$C_{1,4}$ Some visual analysis tasks may demand high cognitive load, causing slow decision processes.
$C_{1,5}$ Algorithmic search shows too many options.
$C_{1,6}$ ...

Similarly, S_2 may be caused by the following shortcomings in P_j or its preceding processes:

$C_{2,1}$ A poorly designed algorithm may cause erroneous outputs.
$C_{2,2}$ An erroneous data pre-processing process may cause the inaccuracy of an algorithm.
$C_{2,3}$ The difference between the statistical properties of the training data and those of the data being analyzed may cause the inaccuracy of a machine-learned algorithm.
$C_{2,4}$ Failing to take into account all variables may cause the inaccuracy of a decision process.
$C_{2,5}$ Poorly tuned parameters may cause the inaccuracy of a parameterized algorithm.
$C_{2,6}$ ...

In addition, $C_{1,2}$ may be antecedently caused by insufficient statistical aggregation, $C_{2,2}$ may be antecedently caused by poor data collection processes, and so forth. From all identified causes, one could start to search for optional remedies for each cause. For example, $C_{1.2}$ may be addressed by the following remedies:

$R_{1,2,1}$ To introduce more statistical aggregation of the data.
$R_{1,2,2}$ To introduce overview first and details on demand.
$R_{1,2,3}$ To introduce algorithmic filtering, clustering, summarization, etc.
$R_{1,2,4}$ To introduce algorithmic support for focus-and-context visualization.
$R_{1,2,5}$ ...

Consequentially, remedies being considered may cause side-effects. For example, it is not difficult to imagine that $R_{1,2,3}$ and $R_{1,2,4}$ could cause side-effects like the symptom S_2. This leads an iterative process for design optimization. As illustrated in Figure 1, one major challenge is that the number of practical instances (i.e., symptoms, causes, remedies, and side-effects) could grow rapidly during the iterative process.

3.2. Theoretical Abstraction

We have considered several schemes that may be useful for categorizing symptoms, causes, remedies, and side-effects, including:

- **Three Components of VA Systems**: *Analysis*, *Visualization*, and *Interaction*.
- **Four Components of VA Systems**: *Statistics*, *Algorithm*, *Visualization*, and *Interaction*.
- **Three Factors for Requirement Analysis**: *Data*, *User*, and *Task*.
- **Four Levels of Visualization Tasks**: *Dissemination*, *Observation*, *Analysis*, and *Model-Development*.
- **Three Information-Theoretic Measures**: *Alphabet Compression*, *Potential Distortion*, and *Cost*.

The main considerations for choosing categorization schemes in this work include the ability and ease (a) to distinguish practical instances, (b) to reason about causal relations, and (c) to have a moderate number of abstract entities in a combined categorization scheme (e.g., <30). The reason for (c) is that given k abstract entities, there are up to $3k^2$ causal relations between symptoms and causes, causes and remedies, and remedies and side-effects.

After careful examination of the above schemes, we have decided on a two-dimensional scheme by combining four components of VA systems and three information-theoretic measures. Readers can find our detailed reasoning in Appendix A. Below we first briefly describe the two dimensions of the combined scheme. We then describe the abstract entities (i.e., categories) in the scheme, and how an abstract entity is instantiated by practical instances.

Four Components of VA Systems. It has been common to decompose a VA system into two major components, *data analysis* (or *data mining*) and *interactive visualization* (or *visual data exploration*) [KAF*08]. We also often see the further uncoupling of *visualization* and *interaction*. Although it may be less common, the further decomposition of *data analysis* into *statistics* and *algorithm* can be conceptually meaningful in reasoning about the interrelations of these components. As Chen et al. articulated in [CTBA*11], for "big data" problems,

- *Statistical methods* alone cannot convey an adequate amount of information for humans to make informed decisions — hence the need for *visualization*;
- *Algorithms* alone cannot encode an adequate amount of human knowledge about relevant concepts, facts, and contexts — hence the need for *interaction*;
- *Visualization* alone cannot effectively manage levels of details about the data or prioritize different information in the data — hence the need for *analysis* and *interaction*;
- Direct *interaction* with data alone is not scalable to the amount of data available — hence the need for *analysis* and *visualization*.

With the decomposition we can observe some major trade-offs, such as between *statistics* and *visualization* and between *algorithms* and *interaction*. The optimization along each pair of components can be considered as finding a sweet spot in answering a design question, e.g., "When do we pause statistics and start visualization?" and "When do we pause algorithmic computation and start interaction?" [CGJ*16]. For example, when dealing with a time series, there would be a sweet spot that optimizes the amount of statistical aggregation before visualization. In a computer-supported decision workflow, there would be a trade-off between the amount of automation (algorithms) and the amount of human involvement (interaction). Those who have had experience of designing a VA system are all aware of the trade-offs between other pairs of components as well as more complicated trade-offs among three or four components.

Three Information-Theoretic Measures. Chen and Golan intro-

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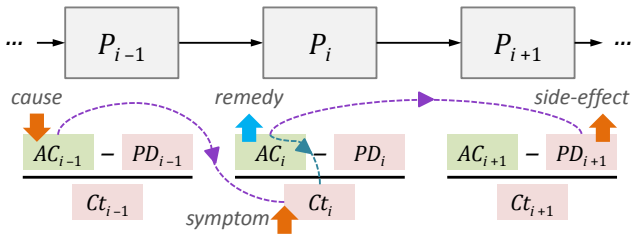


Figure 2: *The cost-benefit of each process can be measured using three fundamental measures,* Alphabet Compression *(AC),* Potential Distortion *(PD), and* Cost *(Ct). The changes of these measures in one process may affect the measures in the subsequent processes.*

duced an information-theoretic metric for measuring the cost-benefit ratio of a VA workflow or any of its component processes [CG16]. The metric consists of three fundamental measures that are abstract representations of a variety of qualitative and quantitative criteria used in practice, including operational requirements (e.g., accuracy, speed, errors, uncertainty, provenance, automation), analytical capability (e.g., filtering, clustering, classification, summarization), cognitive capabilities (e.g., memorization, learning, context-awareness, confidence), and so on. The abstraction results in a metric with the desirable mathematical simplicity [CG16]. In this work, we use the qualitative form of the metric:

$$\frac{\textit{Benefit}}{\textit{Cost}} = \frac{\textit{Alphabet Compression} - \textit{Potential Distortion}}{\textit{Cost}} \quad (1)$$

As illustrated in Figure 2, given three processes P_{i-1}, P_i, P_{i+1}, in a VA workflow. We can examine how each process transforms its input data space to its output data space in terms of the trade-off among the three measures:

- *Alphabet Compression* (AC) measures the amount of entropy reduction (or information loss) achieved by a process. As it was noticed in [CG16], most VA processes (e.g., statistical aggregation, sorting, clustering, visual mapping, and interaction), feature many-to-one mappings from input to output, hence losing information. Although information loss is commonly regarded harmful, it cannot be all bad if it is a general trend of VA workflows. Thus the cost-benefit metric makes AC a positive component.
- *Potential Distortion* (PD) balances the positive nature of AC by measuring the errors typically due to information loss. Instead of measuring mapping errors using some third party metrics, PD measures the potential distortion when one reconstructs inputs from outputs. The measurement takes into account humans' knowledge that can be used to improve the reconstruction processes. For example, given an average mark of 62%, the teacher who taught the class can normally guess the distribution of the marks among the students better than an arbitrary person.
- *Cost* (Ct) of the forward transformation from input to output and the inverse transformation of reconstruction provides a further balancing factor in the cost-benefit metric in addition to the trade-off between AC and PD. In practice, one may measure the cost using *time* or a monetary measurement.

As detailed in [CG16], the AC, PD, and cost of a process P_i affects not only mutually within P_i, but also the *AC*, *PD*, and cost of the subsequent processes, i.e., $P_{i+1}, P_{i+2}, \ldots, P_n$. For example, if P_i offers a high amount AC (e.g., more aggressive statistical aggregation or a lower resolution visualization), it may increases the amount of *PD* at P_i, but reduce the cost of P_{i+1} as it has less input data to handle.

Theoretically, improving a VA system is a global optimization of the VA workflow to be supported by the system, which is unfortunately an intractable problem in most applications that require a reasonably complex VA workflow. This theoretical conclusion about system design is consistent with the challenges in designing VA systems in practice. Nevertheless, the metric allows the designers of VA systems to examine the trade-off at each process of a workflow systematically with only three abstract measures, and the causal effects in subsequent processes.

One may find a more detailed qualitative description and discussion in Appendix A.4 and a mathematical description in [CG16].

The 24 Abstract Entities. Combining the four components and three measures yield 12 categories. In order to reason about problems and solutions in design optimization, we consider two levels of each category, *high* or *low*. In general, "high" means "higher than desirable" in terms of symptoms, causes, and side-effects, or "to make it higher" in terms of remedies, while "low" means "lower than desirable" in terms of symptoms, causes, and side-effects, or "to make it lower" in terms of remedies. Because "low PD" and "low cost" cannot be a bad thing, these two types of abstract entities are only associated with remedies.

The combined categorization scheme thus has 24 abstract entities, namely (statistics, high AC), (statistics, low AC), (statistics, high PD), (statistics, low PD), (statistics, high Ct), (statistics, low Ct), ..., (algorithm, high AC), ..., (visualization, high AC), ..., (interaction, high AC), ..., (interaction, low Ct).

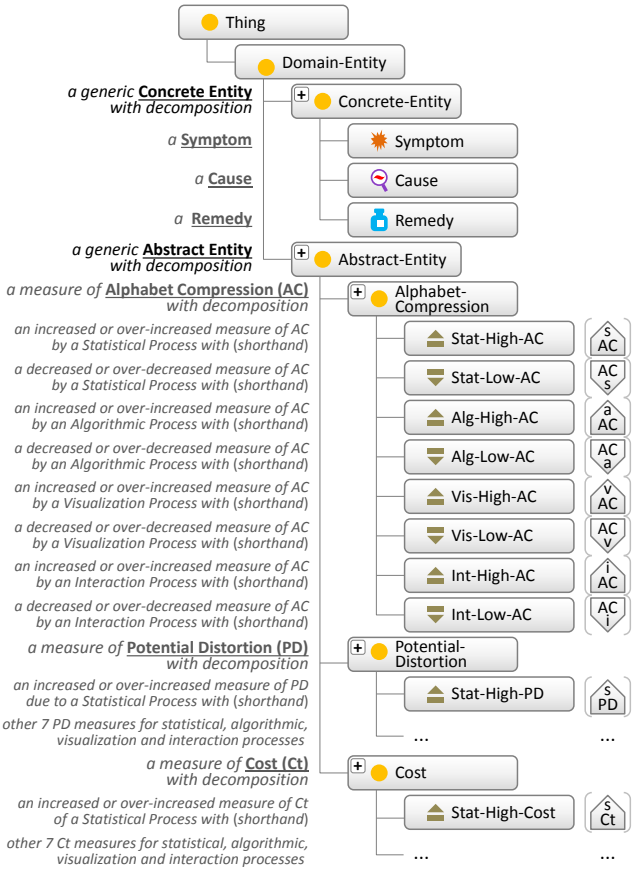
Abstraction and Instantiation. Once we have defined the above categorization scheme, mapping from practical instances to abstract entities is relatively trivial. Using the examples in Section 3.1, we can easily establish the following mappings:

$$\begin{aligned}
S_1 &\longrightarrow \text{(interaction, high Ct) | (visualization, high Ct)}\\
S_2 &\longrightarrow \text{(algorithm, high PD)}\\
C_{1,1} &\longrightarrow \text{(interaction, low AC) | (interaction, high PD)}\\
C_{1,2} &\longrightarrow \text{(visualization, low AC)}\\
&\ldots\\
R_{1,2,1} &\longrightarrow \text{(statistics, high AC)}\\
R_{1,2,2} &\longrightarrow \text{(visualization, high AC) \& (interaction, high AC)}\\
&\ldots
\end{aligned}$$

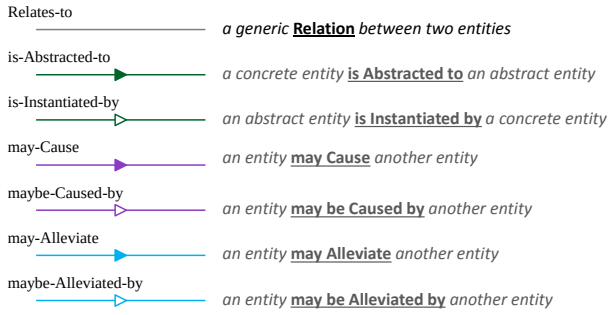
Meanwhile, when we have an abstract notion of a remedy, such as (algorithm, high AC), it is relatively more difficult to instantiate it to some concrete actions. One approach is to build an extensive list of practical instances of optional remedies, and VA designers can browse these options to get some inspiration. Those listed in Appendix B.2 can be a starting point, and the VA community can gradually grow these lists in the future. Meanwhile, for an experienced VA system designer, the abstract notion of remedy (algorithm, high AC) implies a suggestion to increase the amount of entropy reduction using algorithms. Knowing that this is for a specific problem in

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(a) Two major subclasses: Concrete and Abstract entities.



(b) Several types of relations (i.e., properties in OWL) between entities.

Figure 3: *The main components of the ontological framework of IVAS are shown in their visual notations.*

a specific process, the designer can likely figure out what algorithm may offer a solution to the problem.

4. An Ontological Framework for Abstract Reasoning

The term *ontology* comes from the Greek prefix *onto-* (meaning "being" or "that which is") and suffix *-logia* (meaning "logical discourse", "study", or "theory") [Mor63]. In computer science, an ontology is a form of knowledge representation [Sow84], and using ontologies to record knowledge in visualization and visual analytics is part of the theoretical foundation of our field [CGJ*17]. As an ontology for recording problems and solutions in designing visual analytics systems will take years or decades to mature, it is necessary to define an *ontological framework* that can be used to govern the evolution of such an ontology. Here we outline an ontological framework for this purpose, and it is called IVAS (*Improving Visual Analytics Systems*). IVAS consists of a collection of basic definitions and notional conventions and a core structure for supporting abstract reasoning, which are described in the following two subsections. There is a website (URL: https://sites.google.com/view/ivas/) for supporting the evolution of an ontology based on IVAS (see also Appendix B).

4.1. Ontological Definitions and Visual Notations

Sacha et al. [SKKC19] recently developed an ontology VIS4ML as a knowledge base for supporting VA-assisted machine learning workflows. They also outlined an overarching ontological structure for VA processes and workflows based on the baseline schematic representation by Keim et al. [KAF*08]. As IVAS is for supporting the development of VA processes and workflows, we anticipate that VIS4ML, IVAS, and other parts of the overall VA ontology under development will be integrated together in the future. We hence adopted the conventions of VIS4ML for specifying IVAS.

Figure 3(a) shows the class hierarchy of IVAS. The subclass Concrete-Entity encompasses all practical instances of problems and solutions. This subclass is further divided into three sub-subclasses: Symptom, Cause, and Remedy. Readers have already encountered a number of examples of these in Section 3.1, and further examples can be found in Appendix B.2.

- The Symptom entities are records that describe observable problems in a VA workflow, and each Symptom entity S_x can typically be expressed verbally as "One problem here is that $[S_x]$."
- The Cause entities record causal relations, and each Cause entity C_y can typically be written as a statement "Alpha may cause Beta." Beta is usually a generic description of a collection of symptoms, while Alpha can be a less-observable symptom in the process concerned or a symptom in a preceding process.
- The Remedy entities are records that describe actions to be taken, and each Remedy entity R_z is typically in the form of a "to do" statement. Actions can be technical as well as non-technical.

Since side-effects are types of symptoms, we do not define a separate sub-subclass for side-effects.

The subclass Abstract-Entity essentially contains the 24 abstract entities described in Section 3.2. They are grouped into three sub-subclasses corresponding to the three information-theoretic measures. While each of the 24 abstract entities indicates one of the two levels, high or low, the three sub-subclasses Alphabet Compression, Potential Distortion, and Cost can be used in situations where the level indication is not applicable. All 24 abstract entities can be visually denoted using shorthand glyphs.

4.2. Abstract Reasoning

There is little doubt that reasoning with concrete entities is more intuitive and does not require learning new terminologies. However,

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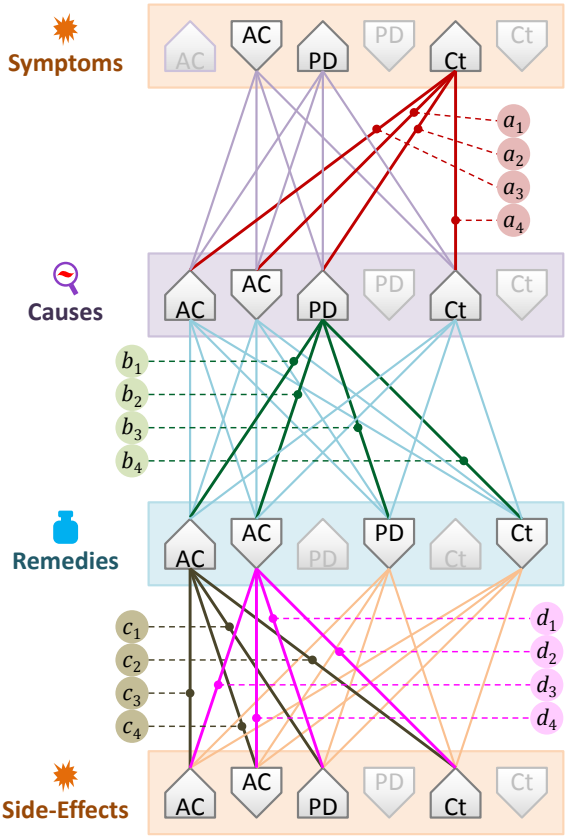


Figure 4: *The possible paths from symptoms, to causes, remedies, and side-effects in abstract reasoning. All paths are of a downward direction, and arrows are omitted for alleviating cluttering.*

because we do not have a mature ontology that provides relatively compressive lists of symptoms, causes, remedies, and side-effects, VA practitioners have to rely on their knowledge and experience to deliberate many different possible causal relations, such as between symptoms and causes, between causes and remedies, and between remedies and side-effects. Although it will take some effort to become accustomed to the 24 abstract entities, the space of abstract reasoning is much smaller, and easily manageable.

Consider the three processes in Figure 2 as an example. If a symptom indicates that the cost of P_i is high, the possible direct causes among these three processes may include: (a_1) insufficient AC by P_i or P_{i-1} (e.g., too much data to handle); (a_2) too much PD due to P_i or P_{i-1} (e.g., the adverse consequences of errors); (a_3) too much AC by P_{i-1} or P_i (e.g., highly abstract visualization leads to high cognitive load); and (a_4) high cost in P_{i-1} or part of P_i leads to further increasing cost in P_i (e.g., a high visualization cost leads to a high interaction cost). These four paths are illustrated in the upper section of Figure 4.

If one finds that the cause is due to the high PD of P_i, the possible remedies may include: (b_1) to increase the AC of P_i or P_{i-1} (e.g., providing overview visualization); (b_2) to decrease the AC of P_i or P_{i-1} (e.g., less aggressive statistical aggregation); (b_3) to decrease the PD of P_i or P_{i-1} (e.g., reducing the errors of the preceding process); and (b_4) to decrease the Cost of P_i (e.g., reducing the cognitive load if P_i is a human-centric process). These four paths are illustrated in the middle section of Figure 4.

Meanwhile, as illustrated in the lower section of Figure 4, one can logically reason about the side-effects. If the remedy is to increase AC in P_i, the potential side-effects may include: (c_1) increasing the PD in P_i or P_{i+1} (e.g., too much information loss may cause errors); (c_2) increasing the Cost of P_i (e.g., a more complex process may increase the cost); (c_3) increasing the AC of P_{i+1} (e.g., information loss leads an over-simplified decision process); and (c_4) decreasing the AC of P_{i+1} (e.g., the provision of overview leads to more options for interaction).

If the remedy is to decrease AC in P_i, the potential side-effects may include: (d_1) increasing the PD in P_i or P_{i+1} if they are human-centric processes (e.g., too much data or too many options may be confusing); (d_2) increasing the Cost of P_i (e.g., shifting one part of the cost to another part) or increasing the Cost of P_{i+1} (e.g., less AC means more output data, which may increase the burden of the succeeding process); (d_3) increasing the AC of P_{i+1} (e.g., forcing the succeeding process to lose more information). (d_4) decreasing the AC of P_{i+1} (e.g., ineffective filtering leads to cluttered visual mapping).

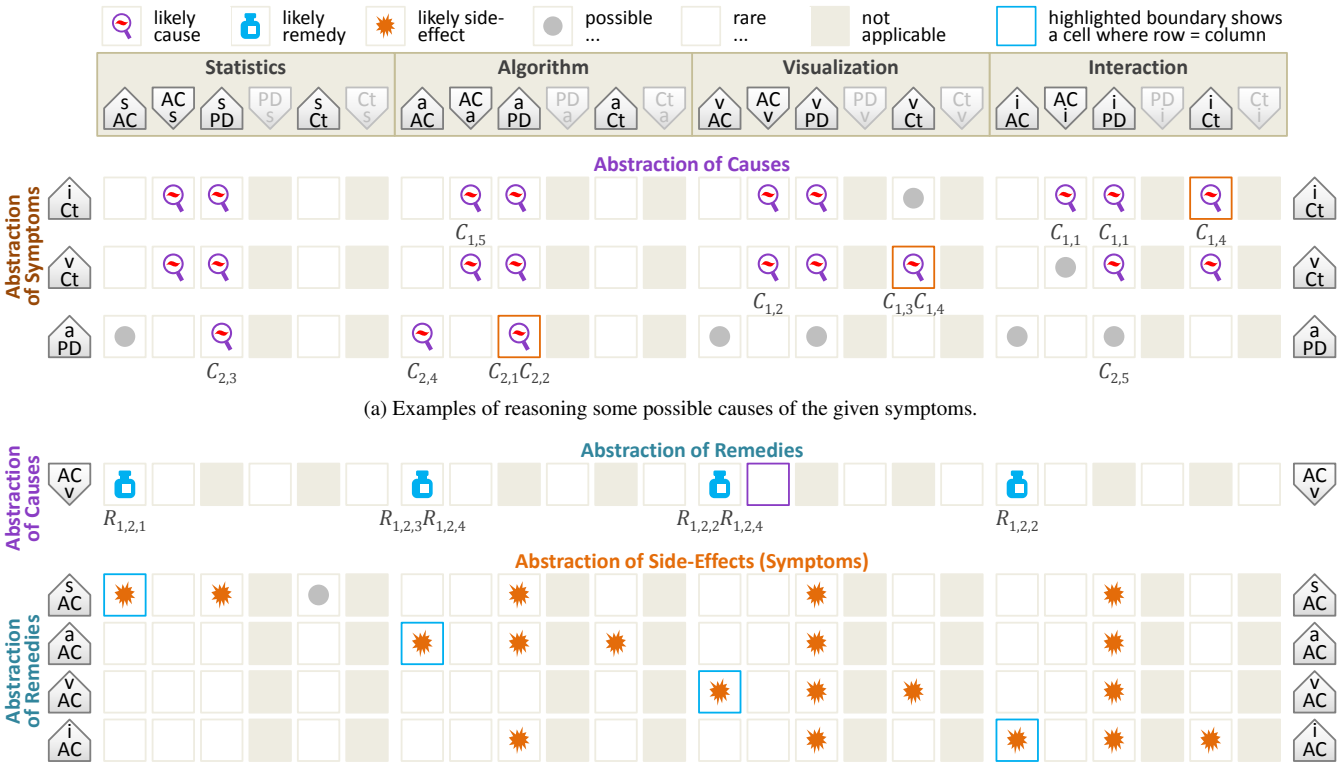
Here we assume that low PD and low Cost are always desirable, and they cannot be symptoms or side-effects. Following the same logic, high PD and high Cost are always undesirable, and they cannot be remedies. However, the reasoning about the amount of AC is more complex. As we discussed before, entropy reduction (e.g., AC) is the general trend of any VA workflow. A process with a higher level of AC passes less output data to the succeeding process, contributing positively towards a more efficient VA workflow. Meanwhile, entropy reduction also implies information loss, which could cause more PD. Hence we always consider both increasing and decreasing AC as possible causes and optional remedies.

All of the above abstract reasoning is based on the cost-benefit metric in Eq. (1) and its quantitative version in [CG16]. More detailed discussions on the relationships among the three measures can be found in Appendix A.4. Meanwhile, as the 24 abstract entities feature another dimension of the four components of VA systems, the relationships among the 24 abstract entities are further constrained by the typical relationships among the four components. In general, it is more likely for an algorithm to use some statistical measures, and less common for statistical analysis to be built on algorithmic results. It is more likely for a visualization process to use algorithmic filtering, and less common for an algorithm to be built on humans' visual analysis. Such inequalities in occurrences should not be generalized to unidirectional dependency. For example, the statistical decisions based on a set of ensemble models would dispute the former inequality, while clustering after crowd sourcing visual analysis would dispute the latter inequality.

Recall the two symptoms S_1 and S_2 in Section 3.1. S_1 is abstracted to entities Int-High-Cost and Vis-Hight-Cost, while S_2 to entity Alg-High-PD. Figure 5(a) shows three rows of possible causes for the three abstract symptoms respectively. As annotated beneath each row, the causes suggested in Section 3.1 instantiate some abstract causes. For those abstract causes without any annotation, it

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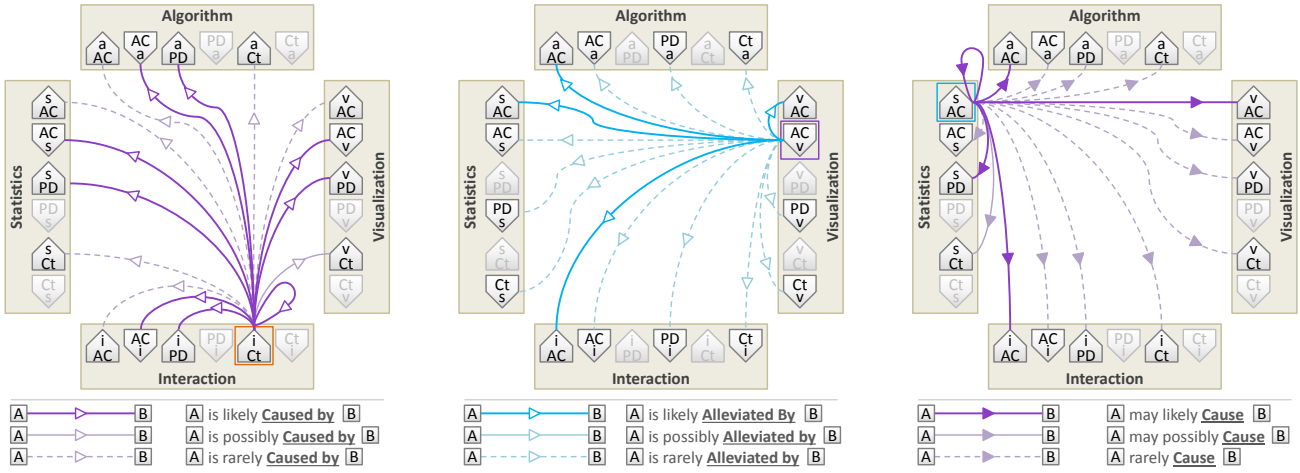
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(a) Examples of reasoning some possible causes of the given symptoms.

(b) Examples of reasoning some optional remedies for a given cause and potential side-effects.

Figure 5: *Using the abstract entities, one can conduct abstract reasoning from symptoms to causes, to remedies, and to side-effects in a scalable manner. The rows in this figure are extracted from three* 24×24 *adjacency matrices in Appendix B.1.*



(a) From symptom Int-High-Ct to causes

(b) From cause Vis-Low-AC to remedies

(c) From remedy Stat-High-AC to side-effects

Figure 6: *The corresponding graph representations for the first row in Figure 5(a) and the first two rows in Figure 5(b).*

is not difficult for a VA practitioner to imagine practical instances that would instantiate such causes in the context of an application.

Having confirmed one or a few causes, a VA practitioner can continue abstract reasoning for remedies. For example, if symptom S_1 "slow decision process" was caused by $C_{1,2}$ "too much data to visualize", i.e., entity Vis-Low-AC, the practitioner can search for remedies using abstract entities as shown in Figure 5(b) with the annotations of the four suggested remedies mentioned in Section 3.1. For each remedy, we can further examine potential side-effects by examining the causal relationship between remedies and side-effects. If a highly desirable remedy has a side-effect causing concerns, the VA practitioner may treat this as a symptom, starting a

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new iteration from symptoms to causes, to remedies, and to side-effects. Figure 1(b) illustrates this iterative reasoning process.

The rows in Figure 5 are extracted from the three 24×24 adjacency matrices in Appendix B.1, which specify the relationships among the 24 abstract entities. Figure 6 shows the node-link diagrams for three rows in Figure 5.

5. Case Studies

There are many stakeholders in the processes of designing and evaluating VA systems. Among them, VA practitioners, who are responsible for designing, improving, and evaluating VA systems and VA workflows, are expected to benefit from the proposed ontological framework and the methodology of abstract reasoning described in the previous two sections. Consider three types of VA practitioners with different levels of involvement in the development life-cycle and different access to the users:

- A **Designer** of a VA system can complement the traditional design methods based on concrete instances with abstract reasoning under the proposed ontological framework. On the one hand, the designer will likely have direct access to the development life-cycle and some users, and will be expected to have firsthand experience of the existing workflow (i.e., the reference system), and in-depth understanding about all of the tasks to be performed and most of the problems encountered by the users. On the other hand, the designer could also be biased by some previous design experience or preconception about the task requirements, leading to some oversight during the design and development.
- A **User Experience (UX) Expert** conducts relatively independent evaluation of a design specification or a newly-developed VA system. The UX expert is expected to have some knowledge about the development life-cycle and direct access to some users. The UX expert can use the methodology of abstract reasoning to reanalyze the symptoms-causes-remedies featured in the design specification or the new VA system, while using all abstract entities for the possible side-effects to frame questions to be examined during in-depth evaluation using empirical methods.
- A **Reviewer** of a technical report or research publication about a newly developed VA system needs to assess any improvements or shortcomings resulting from the new technical development. Such a reviewer usually does not have direct access to either the development life-cycle or users of the VA system. Ideally, a reviewer can be made aware of the abstract reasoning conducted by the system designer and the UX expert, and the factual information about the problems and solutions. The reviewer can then use abstract reasoning to judge the merits and demerits of the new development by analyzing independently the symptoms, causes, remedies, and side-effects, and hence the appropriateness of the chosen solutions.

In this work, we conducted our case studies from the perspective of a reviewer. We examined several application papers in the literature on different VA developments, and used the proposed ontological framework and the methodology of abstract reasoning to analyze the reported VA systems. For each paper, We first produced a report of the symptoms, causes, remedies, and side-effects independently without involving any co-author of the paper. We then asked one of the original co-authors to comments on the report, which implicitly allows us to evaluate the effectiveness of the abstract reasoning. The full reports and the author's feedback are given in Appendix C. On average, it takes about 40-60 minutes to conduct abstract reasoning for each paper, including writing a report of 500-1000 words but excluding the time for the first-reading of the paper. While our independent analysis has confirmed the merits of the solutions described in these papers, the analysis, in some cases, also pointed out several aspects that the original authors did not anticipate during their development.

In general, the case studies from the perspective of a reviewer have confirmed the effectiveness of the proposed ontological framework and the methodology of abstract reasoning. Since a system designer or a UX analyst has much more information about the development life-cycle and the users than an independent reviewer, they can complement their traditional design and evaluation methods by carrying out abstract reasoning as a reviewer. Below we report three case studies.

5.1. VA for Evaluating Operational Performance

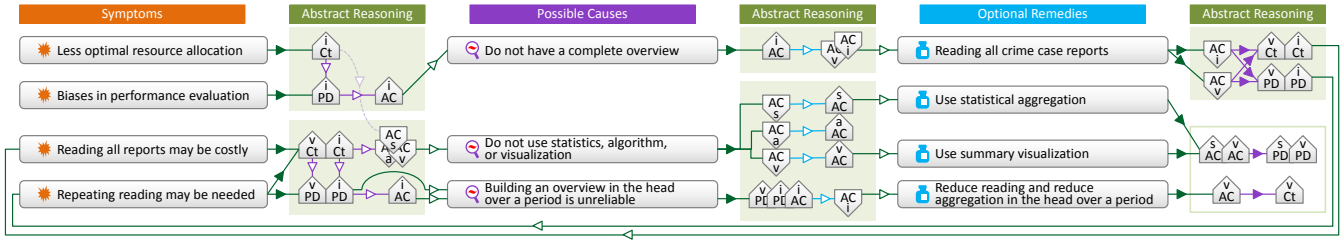
Zhao et al. reported a VA solution designed to improve an existing workflow for supervisors to evaluate the performance of officers in different operations [ZMX*17]. In the existing workflow, which was not supported by VA, supervisors in law enforcement agencies were required to perform resource allocation tasks regularly. There were concerns that supervisors might be biased by their experience of interaction with individual officers. Their knowledge about an officer's effectiveness was limited by the cases that they supervised or knew well. Even for such cases, they might fail to notice some characteristics of the officer's performance. The authors mentioned that "spending time to read every individual crime case report" was an option considered but not adopted.

Figure 7(a) summarizes the abstracting reasoning of the symptoms, causes, remedies, and side-effects in this application. Our independent review of the paper identified two iterations in the development life-cycle. The first two symptoms were identified in the paper. The biases are naturally mapped to abstract entity Int-High-PD, while the ineffective resource allocation mapped to entity Int-High-Cost. The identified cause that supervisors did not have the full access implies, in abstraction, that there is too much information loss, and hence entity Int-High-AC.

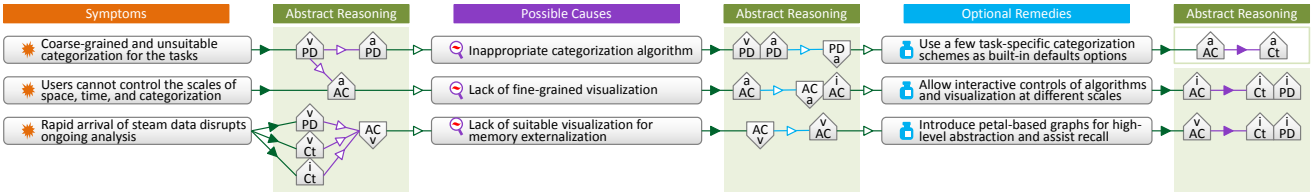
Given the abstract cause Int-High-AC, the Cause-Remedy matrix in Appendix B.1 suggests optional remedies as reducing AC in terms of interaction, while increasing AC through statistics, algorithm, or visualization. The most obvious remedy for Int-High-AC is Int-Low-AC, which led to the suggestion of "reading all reports", i.e., directly interacting with a lot of data. Figure 7(a) also shows Vis-Low-AC as a remedy because "reading all reports" is considered as within the broad scope of visualization. Because no statistics or algorithm has been deployed yet, abstract remedy Int-Low-AC has two potential side-effects, Int-High-Cost and Int-High-PD according to the Remedy-Side-Effect matrix in Appendix B.1. Similarly, Vis-Low-AC also has two potential side-effects. The expensive cost of reading all reports was mentioned in [ZMX*17], while the potential distortion in gaining an impression about officers' performance through reading reports over a period is another critical side-effect.

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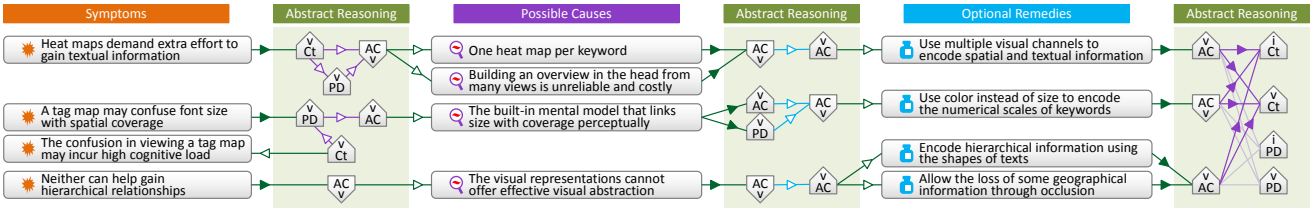
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(a) The two iterations for reasoning about the symptoms, causes, remedies, and side-effects in relation to the MetricsVis system [ZMX*17]



(b) The abstract reasoning about the symptoms, causes, remedies, and side-effects in relation to a VA system for social media analysis [ZAM*16]



(c) The abstract reasoning about two empirical studies for evaluating three visual designs for geospatial text visualization [ZSE*18]

Figure 7: *Application of the proposed ontological framework and the methodology of abstract reasoning to the analysis of problems-solutions in three practical case studies.*

This led to the second iteration, where the potential remedies of Stat-High-AC, Alg-High-AC, and Stat-High-AC were considered. In [ZMX*17], the authors detailed their technical solution involving a process of characterizing case reports using a number of statistical measures, and another process for the supervisors to view summary visualizations of these statistical measures. Although the paper did not mention why the remedy Alg-High-AC was not adopted, an experienced VA reviewer would easily appreciate the difficulties in using an algorithm to assign officers to crime cases.

Using abstract reasoning, a reviewer could also identify and question about some potential side-effects, since using statistics and visualization could lead to potential distortion and extra cost. In hindsight, the authors could discuss further remedies, such as alleviating Stat-High-PD through reading reports on demand, and alleviating Vis-High-PD and Vis-High-Cost through training.

5.2. VA for Microblog Data Analysis

Zhang et al. presented an application where they introduced multi-scale data aggregation and visualization [ZAM*16]. Their main objective was to provide emergency services with a VA system for gaining incident-related information from real-time microblogs. Their pre-development analysis identified a few shortcomings of existing systems, including: (i) the text labels produced by topic modelling and sentiment analysis were not suitable for the visualization tasks; (ii) the categorization was too coarse; (iii) the rapid arrival of the streaming data caused rapid updates of visualization, disrupting the ongoing analysis.

As illustrated in Figure 7(b), the first shortcoming appears to be imprecision in visualization but is caused by inadequacy of the underlying algorithm, i.e., Alg-High-PD. The second shortcoming is a typical problem due to excessive compression by an algorithm, i.e., Alg-High-AC, thus losing too much information. The third shortcoming indicates the symptom of Vis-High-PD due to rapid changes of visual information, and symptoms Vis-High-Cost and Int-High-Cost due to repeatedly going backwards and forwards while remembering relevant information. The cause is the lack of suitable visualization for memory externalization (Vis-Low-AC).

Our abstract reasoning indicates three promising remedies. Two of them were considered and reported in [ZAM*16]. The authors provided an improved VA system that enabled users to select different spatial and temporal scales and different categorical details for data aggregation, addressing the second shortcoming. By shifting some entropy reduction from algorithm to interaction, the system produces summary information that is more relevant to tasks. They also provided petal glyphs (a high-level of alphabet compression Vis-High-AC) to depict multivariate data at different locations, addressing the third shortcoming as the changes of petal glyphs are more noticeable in spatiotemporal visualization.

The authors indicated that the remedy for the second shortcoming also addressed the first shortcoming as the users could now control what text labels are generated. Meanwhile our abstract reasoning suggests that one could consider addressing the first shortcoming directly by developing a few task-related models for topic and sentiment analysis and including them in the system as default ana-

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lytical schemes. Such a feature could also reduce some interaction cost, which may be a side-effect of the fine-grained interaction.

5.3. A Design Study for Geospatial Text Visualization

Zhang et al. conducted two empirical studies to compare three visual designs for geospatial text visualization [ZSE*18]. The main design requirement is to depict textual information at different locations on a map, i.e., data tuples in the form of [x, y, *keywords*, *volume*]. This work can be considered as the optimization of a visualization process within a VA workflow since the preceding analytical algorithms do not change within this optimization exercise.

As shown in Figure 7(c), the authors considered the symptoms if a heat map or tag map visualization were used, and identified a new requirement for visualizing the hierarchy of the geographical region concerned. Our abstract reasoning indicates that there are three visualization tasks for viewing (D_t) the textual labels, (D_h) the hierarchical structure, and (D_g) the geographical map respectively.

The authors considered three visual designs, referred to as M-bd, M-bh, M-sp, featuring different levels of abstraction for D_t, D_h, and D_g. Let $\mathbf{AC}(x)$ denote the amount of alphabet compression.

- For D_t, we have $\mathbf{AC}$(M-bd) > $\mathbf{AC}$(M-bh) > $\mathbf{AC}$(M-sp) in terms of losing useful textual information.
- For D_h, we have $\mathbf{AC}$(M-bd) > $\mathbf{AC}$(M-bh) > $\mathbf{AC}$(M-sp) but most of the information retained by M-bh and M-sp are not useful.
- For D_g, we have $\mathbf{AC}$(M-bd) < $\mathbf{AC}$(M-bh) < $\mathbf{AC}$(M-sp) in terms of losing information due to occlusion.

In addition, for D_t the perception of text labels with M-bd (on the boundary) is more difficult than that with M-bh and M-sp; and for D_h the perception of the spatial boundary is relatively the most difficult with M-bh.

The results of the two empirical studies in [ZSE*18] are consistent with the above abstract reasoning. The time cost for visualization may increase because of losing useful information and high potential distortion, while it may reduce because of losing less useful information. Here we use $\mathbf{Ct}(x)$ to denote the cost of each method. The first empirical study focused on the perception of D_t, and it showed that the time costs are $\mathbf{Ct}$(M-bd) > $\mathbf{Ct}$(M-bh) > $\mathbf{Ct}$(M-sp). The second empirical study focused on D_h and D_g, and it showed that the time costs are $\mathbf{Ct}$(M-bd) < $\mathbf{Ct}$(M-sp) < $\mathbf{Ct}$(M-bh).

The abstract reasoning can easily be extended to other visual designs, for example, a variation of M-bd with text labels only on the boundaries or a variation of M-bh with partially translucent text labels to avoid jagged boundaries. We can also use abstract reasoning before applying the conclusions of such empirical studies to practical applications, for instance, for considering additional variables, such as the users' knowledge about the area concerned and the spatial hierarchy in the area. The effort and time required could be less than what the empirical studies suggested, enabling more design iterations and efficient evaluation.

Meanwhile, empirical studies allow us to test hypotheses with performance data. However, they are costly to run, and usually have to fix many variables that would appear in real applications. Hence, abstract reasoning and empirical studies are complementary.

6. Discussions and Conclusions

In this paper, we have presented an ontological framework that can be used as the core for building a comprehensive ontology to support the design, evaluation, and improvement of VA systems. It utilizes an information-theoretic metric to reason about the causal relations between symptoms and causes, causes and remedies, and remedies and side-effects. Once it is learned, the abstract reasoning based on the 24 abstract entities can be conducted quickly, and can be used in conjunction with traditional design methodologies.

The three case studies in Section 5 show that the abstract reasoning can be used to reason about the design options for improving a VA system as well as an individual visualization process. As the third case study shows that abstract reasoning can work in parallel with an evaluation study, helping formulate hypotheses, interpret results, and translate the conclusions to real world applications. As pointed out by Weber et al. [WHC*17], while applications have an indispensable role in visualization and VA, there is also a need of "building deep knowledge". This work is a step towards building deep knowledge through applications. It indicates the potential for setting up a framework, under which authors can present and reviewers evaluate systems on a common ground.

We appreciate that the proposed ontological framework is only an early step towards building a more comprehensive ontology that records many more concrete entities of symptoms, causes, remedies, and side-effects in designing, evaluating, and improving real-world VA systems. With a large collection of real world instances, VA researchers and practitioners will be presented with new challenges and opportunities, very much in the same way as what medicine and healthcare have encountered for centuries and millennia. These will include compiling handbooks (cf. handbooks for clinicians [WRW*17, JRW*18, TO17, KE16, IDF18]), developing regimented approaches for working with domain experts (cf. The Calgary-Cambridge Guide to the Medical Interview [KSD98, SKD98]), using VA to support the exploration of the problem-solution space, and providing VA-assisted decision support in the design, development, evaluation, and optimization of VA processes, systems, and workflows (cf. clinical decision support systems [GAM*05, MKL*14]).

The main limitation of abstract reasoning is that it is based on an information-theoretic metric, which is unfamiliar to most VA practitioners at the moment. There will be some effort required to become accustomed to the reasoning about the trade-offs among the three fundamental measures. Nevertheless, just like almost all historical methodologies based on mathematical and conceptual abstraction, such effort is unavoidable. Since abstraction is an integral part of VA, the development of VA systems must be able to benefit from abstraction, while VA itself can potential enable such abstraction more assessable to VA practitioners.

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